

Tornado geometry and reported loss: a bounded negative industrial result

Path geometry and contextual variables improve broad loss-estimate prediction over simple controls, but errors remain very large and stronger flexible controls beat the selected compact equation. No reliable industrial damage law is demonstrated.

Data and evaluation

2025 NOAA tornado segments with explicit damage estimates, positive path length/maximum width and valid timing: 1072 development/224 confirmation segments in 363/94 complete episode groups. Reported zero damage remains zero; missing damage is excluded, never imputed. Source EF damage-derived ratings and narratives are forbidden predictors.

Target: **Log transformed reported tornado property-damage estimate**, measured as $\ln(1+\text{property damage in USD})$. Primary error: natural-log dollars offset 1 USD. Post-event path and timing diagnostic of reported loss. EF damage-derived scale, casualty counts, narratives, monetary/crop losses excluded from predictors.

NOAA EPISODE_ID outbreak groups; nearby episodes may still be correlated or linked across offices. Not individual property samples. Whole EPISODE_ID groups held; no held-event monetary outcomes. Linked tornado segments within episode remain together.

Findings and equations

$$\hat{y} = \text{clip}(a + b_L \ln(1 + L) + b_W \ln(1 + W) + b_\phi \phi + b_s \sin \theta + b_c \cos \theta, 0, 50)$$

Here $\theta = 2\pi \text{ month}/12$ and ϕ is latitude. The notation and all fitted coefficients are defined below.

P100-018-F01 - reproduction (partially_supported). Recorded path geometry carries descriptive information about broad reported property losses.

The selected robust log-loss equation and its coefficients are given below.

Dimensions and regional/seasonal composition correlate with losses but also with reporting and asset exposure; maximum width times length is only a bounding proxy.

Falsifying evidence and limits: Flexible controls outperform, very large log errors and worse extreme episodes remain. No wind/asset exposure or independent insured loss truth.

P100-018-F02 - informative_falsification (supported). The available geometry does not demonstrate an accurate transferable industrial damage law.

Duration and context ablations 005/006, plus matched robust controls, challenge the proposed law.

Negative duration slope conditional on geometry and strong fit-loss dependence prevent a simple energy/exposure time interpretation.

Falsifying evidence and limits: Reported monetary losses are broad estimates, not direct mechanical response; held worst error and stronger flexible control reject deployment claim.

Selected executable model: attempt_005. $y = \ln(1 + \text{damageUSD})$, predicted $y = \text{clip}(a + b_L \ln(1 + \text{length_km}) + b_W \ln(1 + \text{width_m}) + b_{\text{lat}} \text{latitude} + b_{\text{sin}} \sin(2\pi \text{ month}/12) + b_{\text{cos}} \cos(2\pi \text{ month}/12), 0, 50)$. Robust train-only residual fitting; coefficients=[3.099796961781409, 1.0765641714752674, 1.5023316079008489, -0.1478989662476028, 0.8455297618332634, -0.00746860537787219]. The length/maximum width proxy is not actual exposed assets, swept damage area or kinetic energy.

All coefficients, input definitions, training groups and transformations are in `rules.json`; the runnable reference performs no fitting.

Compact numerical evidence

Errors are computed within declared complete groups (with final survey weights when supplied), then averaged equally across groups.

Frozen model	Development error	Confirmation error	Worst confirmation group
reference	4.284359	4.275538	12.86898
constant	5.490342	5.439115	7.487723
domain	4.978277	5.02809	15.14759
flexible	4.589906	4.243949	9.299377

Postfreeze diagnostic controls were fitted on development data after original confirmation exposure, without reselection. They address estimation-loss asymmetry; they are not fresh preregistered confirmation.

Added control	Development error	Exposed-cohort error
postfreeze_robust_domain	4.920522	4.978022
postfreeze_robust_flexible	4.433188	4.05943

Confirmation contains 224 scored observations in 94 groups. Individual-group scores and every attempted model remain available. Selected 005 confirmation MAE 4.275538 log units versus constant 5.439115, area baseline 5.028090 and flexible 4.243949. Worst selected episode 12.868983 is far worse than constant 7.487723. Postfreeze robust flexible diagnostic improves to 4.059430; it was fitted on development only after first confirmation and was not promoted. Huge log errors defeat precise financial-loss use.

Interpretation, limitations and use

2025 tornado segments with nonmissing explicit damage. Broad estimates including reported 0, not insured claim totals. Maximum width times length is bounding footprint proxy, not actual damaged area. Exposure/building stock and wind speed unavailable; no prospective loss or causal damage law. Whole-group errors/ranges; no row bootstrap or manufactured independent-population confidence. Confirmation becomes exposed after freeze.

Provides a realistic benchmark for scientific abstention: incomplete exposure and wind information cannot be replaced by a visually plausible damage-scaling formula. The practical result is a reproducible demonstration of limited predictability, not a pricing/insurance tool.

Reproducibility and prior art

Run `python run.py` to verify hashes and replay every frozen equation. Predictions use declared input columns only. The shared benchmark evaluator can score alternative equations under the same task; numerical agreement with these coefficients is not required. Public source-aware confirmation is now exposed. No independent experimental replication or certified novelty is claimed.

- Property/crop damage are broad estimates. (checked 2026-10-02).
- Path length in miles and maximum width in yards; multi-segment tornadoes reported in segments. Geometry-loss scaling is a known descriptive approach; absent exposure forbids mechanistic damage claims. (checked 2026-10-02).